INTEGRATING ERC-20 ALLOWANCE EDGES INTO PAGERANK FOR ENHANCED ON-CHAIN WALLET REPUTATION SCORING

CHAPTER 2: Literature Review and Theoretical Framework

by

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2 Literature Review and Theoretical Framework

The research problem set out in Chapter 1 is that transfer-graph reputation scores give no role to spending authorization, an explicit act of trust that every ERC-20 token records, and that their graphs grow with every payment. The objectives O1–O5 and research questions RQ1–RQ5 follow from it. To compare EndorseRank with Adaptive Weighted PageRank (AWP) on this problem, we run each method's graph through PageRank and check the resulting rankings with rank correlations. The gap that remains is taken up in Chapters 3 and 4.

2.1 Blockchain and DeFi foundations

A public blockchain records transactions in hashed blocks, and each block is secured by the ones added after it (Nakamoto, 2008). Once a majority of the participating nodes has accepted a block, rewriting it is costly. That cost allows parties to settle on the ledger without each of them needing a known legal identity (*trust-minimized* settlement).

Bitcoin built a peer-to-peer electronic cash network on this kind of ledger (Nakamoto, 2008). Ethereum keeps the append-only ledger but lets an account hold a program alongside its balance. A transaction can then call one of the program's functions and change the contract's state (Buterin, 2014), and these functions run on the Ethereum Virtual Machine (EVM) (Wood, 2025). Programs stored on the ledger in this way are called smart contracts (Szabo, 1997); any honest node that replays a function call gets the same output (Wood, 2025). As a result, collateralization ratios, liquidation conditions and fee schedules can be enforced by the contract instead of by a trusted third party (Schär, 2021).

Collectively, these contracts make up Decentralized Finance (DeFi). Between them they do the work of exchanges, lending platforms, derivatives markets and asset managers; Werner et al. (2022)

define the protocol classes, and Schär (2021) describes how protocols are layered on one another. Because one protocol can interact with another, new financial instruments can be composed from existing ones (Kitzler et al., 2023). The same property lets a failure spread across several protocols. Successful attacks have included flash-loan exploits (Qin et al., 2021) and price-oracle manipulation (Zhou et al., 2023). Transaction histories on these chains are public, yet the addresses in them are pseudonymous, which makes the identity checks of conventional finance hard to carry out (D. Lin et al., 2025).

2.1.1 Pseudonymity and the credit problem

Conventional credit scoring ties a credit history to a legal entity. On a public blockchain the basic unit of analysis is instead an address (a wallet), which may stand for a legal entity, a trust, an algorithmic agent or a smart contract. One entity can own several addresses (Douceur, 2002), and statistical clustering can recover ownership only in part (Victor, 2020). On-chain lending therefore requires borrowers to post collateral worth more than the amount they borrow; if the collateral value falls below a set threshold, a liquidator may repay part of the debt and receive the corresponding collateral at a discount (Bartoletti et al., 2021). The arrangement is secure, but it ties up capital. An on-chain credit score tries to quantify from on-chain behavior alone how likely an actor is to repay. Recent proposals draw on prospect theory (Hassija et al., 2020), on machine learning applied to lending data (Kotb, 2023; Palaiokrassas et al., 2024), and on models of risk spreading through account networks (D. Lin et al., 2025).

2.2 Ethereum, Layer-2 rollups, and Arbitrum One

Most DeFi applications have been deployed on the Ethereum mainnet, historically the main venue for DeFi innovation and liquidity, but while Ethereum ran on proof of work, it could not raise its block size or shorten its block interval far without weakening its security (Gervais et al., 2016), so growing use of the mainnet raised gas costs and constrained throughput. Layer-2 (L2) proto-

cols try to keep Ethereum's security guarantees while making transactions cheaper. Some execute transactions off-chain, while others batch them and post a succinct summary of the execution to the Ethereum mainnet (Gudgeon, Moreno-Sanchez, et al., 2020). Thibault et al. (2022) survey rollups and compare their implementations. Optimistic L2s such as Arbitrum treat each submitted batch of transactions as valid unless evidence against it is presented within a dispute period (Kalodner et al., 2018).

Arbitrum One is an optimistic L2 of this kind and can execute EVM-based smart contracts. It has become a popular platform for DeFi applications, among them GMX, an exchange for perpetual swaps (GMX, n.d.). We use Arbitrum One because the public BigQuery dataset includes both its ERC-20 `Approval` and `Transfer` events, and because its fees are low enough that the observation window contains dense activity. We make no claim that our results generalize to other L2s or to Ethereum mainnet without further validation. The framework can be applied to any chain that emits the same logs.

2.3 ERC-20 tokens and the allowance lifecycle

The ERC-20 standard defines a common interface for fungible tokens, made up of `transfer`, `approve`, `transferFrom`, `allowance` and the associated events (Vogelsteller & Buterin, 2015). After the owner calls `approve(spender,amount)`, the token contract records that the spender may move up to `amount` out of the owner's balance. The spender can then call `transferFrom` to carry out a token transfer. With EIP-2612 `permit`, an allowance can be updated through an off-chain signature that settles in a single transaction (Lundfall, 2020). The semantics of authorization are unchanged. Figure 2.1 traces an ERC-20 allowance through its lifecycle. An allowance is an authorization and involves no actual transfer. Token owners can grant allowances to routers, vaults or trading contracts that they trust to act within set limits. Revoking (approving zero) and updating (setting a new non-zero allowance) both change the allowance currently in force, and `transferFrom` reduces it; some tokens record the remainder in a new `Approval` event. EndorseRank keeps, for each owner, spender

and token, only the most recent approval, so a newer allowance replaces the previous one outright, and the surviving approval loses weight as it ages even while it stays in force (Section 2.7).

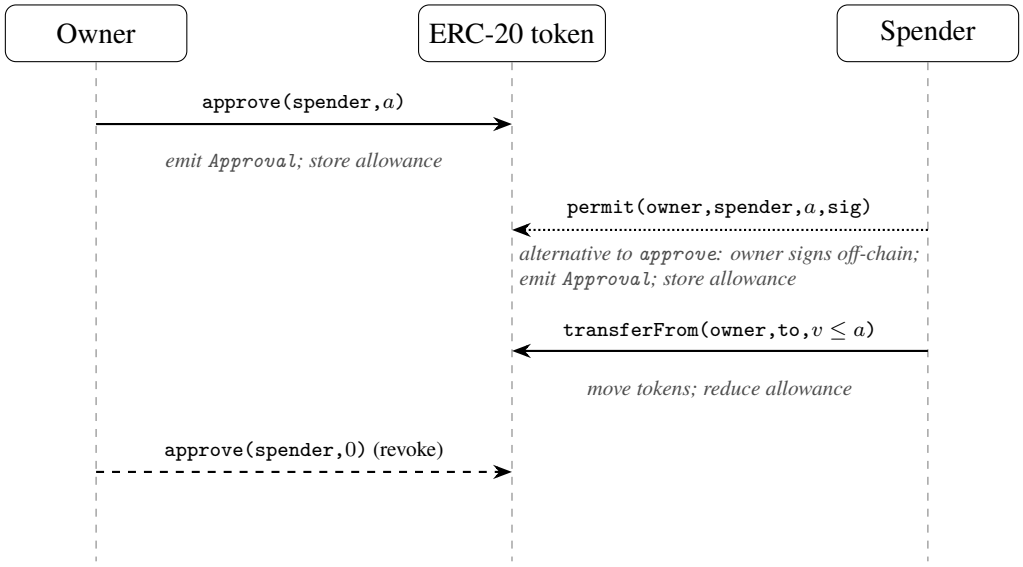


Figure 2.1: ERC-20 allowance lifecycle: `approve`, or its signature-based alternative `permit` (dotted), an optional `transferFrom`, and revoke. EndorseRank keeps the most recent `Approval` event of each owner, spender and token in the observation window and weights it by its age and US-dollar amount; a later approval replaces an earlier one.

In the available data, allowances are sparse compared with transfers. A single approval can cover many transfers, and many wallets have never granted an approval to any external account. The sparsity lowers the cost per iteration, and it also marks a semantic difference, since EndorseRank quantifies authorization to spend whereas a transfer graph records payment flows.

2.4 Graph-based on-chain reputation

PageRank and its variants rank nodes by the influence that flows to them along directed edges (Page et al., 1999). Two recent studies apply this approach to blockchain transaction graphs. H.-D. Do and Do (2023) run PageRank and HodgeRank on Ethereum transactions as a measure of social credit, meant for people who lack a credit history at financial institutions. T. Do et al. (2023) propose Adaptive Weighted PageRank (AWP), the transfer-graph method that this thesis evaluates

beside EndorseRank (Section 2.18). D. Lin et al. (2025) treat the risk rating of Ethereum accounts as a node-ranking problem and propagate risk through the transaction network, so that accounts without labels are rated as well, and Wu et al. (2022) detect phishing addresses from a network embedding of the transaction graph that takes the amounts and times of transactions into account. Bellini et al. (2020) survey blockchain-based trust and reputation management, and Almasoud et al. (2020) review reputation systems built on smart contracts, in which users rate how trustworthy the providers of online services or products are. Data mining on cryptocurrency transactions likewise works from transaction histories, for address linking and for the study of user behavior (Liu et al., 2021). Packin and Lev-Aretz (2024) warn that decentralized credit scores can turn biased decisions into automated processes that outsiders cannot inspect.

Outside the research literature, trust in deployed DeFi and web3 systems is usually established another way. An address builds standing through activity that anyone can inspect on the ledger, and tooling bound to the address summarizes it. One such tool, Human Passport (formerly Gitcoin Passport), collects verifiable credentials, called stamps, for checks such as government identity, biometrics, vouching and earlier web2 and web3 activity. Each stamp carries a weight that the documentation describes as the strength of its human signal, and an address passes when the weighted sum reaches a threshold (Human Passport, n.d.). According to the same documentation, Passport also classifies EVM addresses on several networks, Arbitrum among them, as human or Sybil from their on-chain activity (Human Passport, n.d.). Its purpose is Sybil resistance, so it combines on-chain with off-chain evidence and works as a gate and not as a graded ordering. A review of proof-of-personhood protocols describes how they resist Sybil attacks with subjective inputs such as vouching and voting in place of attestations from central issuers (Siddarth et al., 2020), and a recent review of Sybil attacks on blockchains traces airdrop farming on layer-2 networks such as Arbitrum and assesses proof-of-personhood systems among the defenses (Bordeianu & Popescu, 2026). Such tools answer whether an address is controlled by a distinct person. EndorseRank answers a different question, which addresses other addresses authorize to spend their tokens, and it orders them from on-chain records alone (Table 2.1).

Graph-based methods are attractive because they (i) use only public information, (ii) produce continuous scores that suit ranking, and (iii) admit complexity analysis by number of edges. They are, however, vulnerable to Sybil attacks (Douceur, 2002) and wash trading, and they suffer from the mismatch between graph centrality and actual default risk; PageRank scores are not credit scores in the regulatory sense (Section 2.8).

A propagated score must also beat a simpler one. Where degree correlations are weak, as on the Web graph, PageRank is on average proportional to in-degree and can be approximated from it (Fortunato et al., 2008). New links concentrate on nodes that already have many, as preferential attachment assumes (Barabási & Albert, 1999) and measured network growth shows (Newman, 2001), so link prediction, on Ethereum transaction networks too (D. Lin et al., 2022), uses degree-based predictors as standard baselines (Liben-Nowell & Kleinberg, 2007).

2.5 PageRank formalism

Consider a directed graph $G = (V, E)$ with $|V|$ nodes, and write $w_{ij} > 0$ for the weight of the edge from i to j. That weight stands for influence from i to j, in either the endorsement or the transfer sense introduced above. Let W be the corresponding weighted adjacency matrix, and let $o_i = \sum_j w_{ij}$ be the out-strength of node i. The row-stochastic transition matrix P is then

$$P_{ij} = \begin{cases} w_{ij}/o_i & \text{if } o_i > 0, \\ 1/|V| & \text{if } o_i = 0 \end{cases} \tag{2.1}$$

(dangling nodes spread their mass evenly). Given a damping parameter $d \in (0, 1)$, the Google matrix is

$$G_{\text{PR}} = d\, P^{\top} + \frac{1-d}{|V|}\, \mathbf{1}\mathbf{1}^{\top}, \tag{2.2}$$

and the PageRank vector $\mathbf{r}$ satisfies $\mathbf{r} = G_{\text{PR}}\mathbf{r}$ with $\mathbf{r} \geq 0$ and $\sum_i r_i = 1$ (Langville & Meyer, 2006; Page et al., 1999). Element by element, with the dangling rows of P written as a separate middle

term, this becomes

$$r_j = d \sum_{i:(i\to j)\in E} r_i \frac{w_{ij}}{o_i} + \frac{d}{|V|} \sum_{i:\, o_i=0} r_i + \frac{1-d}{|V|}. \tag{2.3}$$

2.5.1 Random-surfer interpretation

Equation (2.3) can be read probabilistically. At each step a surfer at a node with out-edges follows one of them with probability d, choosing it in proportion to its weight, or teleports to a uniformly chosen node with probability $1 - d$ (Figure 2.2). At a node without out-edges the surfer always jumps to a uniformly chosen node; this is the dangling rule of Equation (2.1). The stationary distribution then ranks nodes by how often they are visited in the long run. Damping keeps rank sinks (groups of nodes with no edge leaving them) from absorbing all the mass. With $d < 1$ and uniform teleportation, every entry of G_{PR} is positive, so G_{PR} is primitive and its Perron vector is unique (Langville & Meyer, 2006). Appendix 8.3.1 gives the argument.

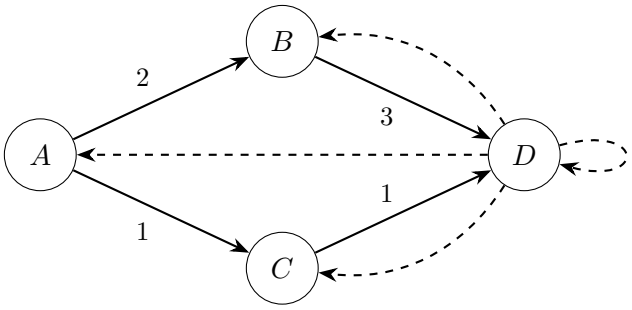


Figure 2.2: Random-surfer intuition for PageRank on the four-node graph of Section 2.13; edge labels are the allowance weights. Solid arrows are graph transitions: with total probability d the surfer follows one of them, chosen in proportion to its weight. Dashed arrows show uniform teleportation, which reaches every node (the current one included) with probability $(1 - d)/|V|$; the dashed fan is drawn only from D, although every node has one. D has no outgoing edge, so under the dangling rule of Equation (2.1) a surfer at D always jumps to a uniformly chosen node, each with probability $1/|V| = 0.25$. That rule, and not damping, keeps the mass that reaches D in the chain.

2.5.2 Power iteration and complexity

In practice, $\mathbf{r}$ is usually obtained by power iteration from the uniform vector $\mathbf{r}^{(0)} = \mathbf{1}/|V|$,

$$\mathbf{r}^{(t+1)} = G_{\mathrm{PR}}\mathbf{r}^{(t)}, \tag{2.4}$$

until $\|\mathbf{r}^{(t+1)} - \mathbf{r}^{(t)}\|_1 < \varepsilon$ or until the iteration cap $I_{\max}$ is reached. On a sparse graph, each iteration costs $O(|E|)$ arithmetic operations (Leskovec et al., 2020). A method that generates fewer edges, for instance by using latest-allowance snapshots instead of full transfer histories, therefore lowers the cost per iteration and sometimes the memory required.

By default this work sets damping to the conventional $d = 0.85$ (Page et al., 1999) and uses two study-specific solver settings, a convergence tolerance $\varepsilon = 10^{-8}$ and a maximum of $I_{\max} = 300$ iterations.[1] Damping is varied over $d \in \{0.75, 0.85, 0.95\}$ in the robustness section.

2.6 Reputation systems lineage

PageRank belongs to a wider family of spectral and propagation-based reputation mechanisms. Kleinberg's HITS algorithm assigns each page two scores, one as a hub and one as an authority (Kleinberg, 1999). EigenTrust extends the idea to peer-to-peer networks and normalizes local trust, so a malicious node cannot raise its own score at will (Kamvar et al., 2003). To suppress web spam, TrustRank starts from a seed set of trusted nodes and propagates their trust outward (Gyöngyi et al., 2004). Reputation is relational in all of these methods. Where trust flows only from a vetted seed set, as in TrustRank, a newcomer cannot raise its score without endorsements from nodes that already hold reputation. With uniform teleportation, which EndorseRank uses (Section 3.6.2), that protection is weaker. Gyöngyi and Garcia-Molina (2005) derive the link farms and farm alliances that most raise a target's PageRank, and Cheng and Friedman (2005) show that symmetric reputation functions such as PageRank can be manipulated with Sybils, whereas some asymmetric functions

[1] The tolerance and the iteration cap were chosen to keep run-times reproducible; they are not put forward as part of the classical PageRank formulation.

based on flows or paths from a trusted node resist them. The Sybil cost model of Section 5.6 prices such a farm on an allowance graph.

On-chain, the Sybil attack remains a central concern (Bordeianu & Popescu, 2026). New wallets cost little to create, whereas wallets that receive allowances or transfers from distinct counterparties cost more. The Sybil-adjusted stability proxies used in this work (inbound counterparty ratio, tenure, active months) partly capture that diversity of interaction, but they do not prevent a malicious actor from manipulating them.

Machine-learning credit models fitted to DeFi lending data (Kotb, 2023) learn from labeled credit outcomes, and an ensemble of fitted models is harder to audit than a fixed rule. EndorseRank and AWP keep to fixed propagation rules that anyone can recompute from the public logs.

2.7 ERC-20 allowances as endorsement

Through the ERC-20 `approve` function (Vogelsteller & Buterin, 2015) and EIP-2612 `permit` (Lundfall, 2020), users state explicitly how much a spender may move on their behalf. EndorseRank reads the latest allowance from an owner to a spender as one endorsement. On each of the five tokens of the study (Section 3.5.1), the latest in-window approval of the pair contributes its time weight times the bounded value weight of its US-dollar amount, as AWP weights a transfer; a pair whose latest value is zero (a revocation or, for tokens whose `transferFrom` emits `Approval`, an allowance spent in full) is dropped, and the terms are summed over tokens (Equation (3.7) in Section 3.6.2). PageRank on the resulting graph G_{approve}, which is sparser than the transfer graph over the same contracts and window, yields the EndorseRank scores $\mathbf{s}_{\text{ER}}$.

Treating allowances as endorsements is a deliberate choice, and an imprecise one. Users approve routers because it is convenient, not necessarily to make a social statement about whether the router is trustworthy. An allowance manager such as Uniswap's Permit2, which owners approve once so that applications can spend through it, collects the endorsements meant for those applications (Section 3.6.2). Contract addresses and EOAs coexist on the owner side as well. Even so, an

allowance signals approval more explicitly than a received transfer, which could come from market making, wash trading (Cong et al., 2023), airdrop farming (Luo et al., 2025) or routing without any intent to endorse.

Approvals have been read as a relation between addresses before: Victor's self-authorization heuristic uses ERC-20 approvals to cluster addresses that it attributes to one entity (Victor, 2020). That reading links addresses and ranks nothing. It is also a caution, since an approval between two addresses of one entity is a self-endorsement, the edge that the Sybil cost model of Section 5.6 prices at gas alone.

2.8 What this research does not claim

Trading outcomes enter this thesis only out of window: in an exploratory run on the GMX V2 contract traders of W0 (Section 4.6.1), as the liquidation-free close rate of the registered window W1 (Section 3.7.2), and in a comparison of the two edge types (Section 4.6.3). That comparison needs an outcome built from neither edge type. Construct validity expects a score to agree with checks of the construct it was built to measure (Cronbach & Meehl, 1955; Messick, 1995), so each edge type's lead on its own checks is partly built in; the GMX V2 trader outcomes give neither score that head start, and for counterparty-risk reputation the relevant one is forced liquidation. No score in this thesis is offered as a predictor of trading profit, loss avoidance or liquidation, or as a measure of creditworthiness or default risk. No score is a security audit of a contract's code either: a contract can contain malicious functions whatever its standing among owners (Z. Lin et al., 2024), and the scores read neither code nor upgrade permissions.

2.9 Rank correlation theory

Reputation scores and the proxies of the construct checks are ordered quantities, and neither has a natural boundary for classification. Following the AWP validation design (T. Do et al., 2023) and classical psychometrics (Cronbach & Meehl, 1955), this research measures two kinds of rank agree-

ment, monotonic and pairwise. Bootstrap confidence intervals quantify the sampling uncertainty of every coefficient of the main analysis (Efron, 1979).

Spearman's rank correlation (Spearman, 1904) is the Pearson correlation of the ranks. For paired ranks (R_i, S_i) of n items without ties, it reduces to

$$\rho = 1 - \frac{6 \sum_{i=1}^{n} (R_i - S_i)^2}{n(n^2 - 1)}. \tag{2.5}$$

This no-ties form does not hold when ties occur. Spearman's ρ is then the Pearson correlation of average ranks (mid-ranks), in which tied items share the mean of the positions they occupy, and every reported ρ is computed this way. In both forms ρ measures monotonic association.

Kendall's τ (Kendall, 1938) rests on counts of concordant and discordant pairs. Of the $n_0 = n(n-1)/2$ pairs (i, j) with $i < j$, n_c are concordant (both rankings order them the same way) and n_d are discordant (the rankings order them oppositely); a pair tied in either ranking is neither. Let $n_1 = \sum_k t_k(t_k - 1)/2$, where t_k is the size of the k-th group of tied values in the first ranking, and define n_2 in the same way for the second ranking. The τ-b variant is then

$$\tau_b = \frac{n_c - n_d}{\sqrt{(n_0 - n_1)(n_0 - n_2)}}, \tag{2.6}$$

and without the tie corrections it becomes $\tau_a = (n_c - n_d)/n_0$. Every τ reported in this thesis is τ_b, computed on the raw scores and proxies. The difference matters for heavily tied scores, such as the raw degrees, which are integer counts; EndorseRank on the matched cohort leaves few ties (Section 4.4). For agreement about pairwise orderings, Kendall's τ is usually the easier coefficient to interpret, and Chapter 4 uses it as the main family-level summary. Its values are compared with those of other methods on the same cohort; no uncited absolute benchmark is used.

2.10 Theoretical framework

This study compares EndorseRank and AWP on reputation structure (H1), construct alignment (H2) and computational efficiency (H3), and then on prediction out of window (RQ4) and on coupling the two layers (RQ5). Three bodies of theory frame the first part: hyperlink–citation propagation, construct (domain) alignment, and computational complexity. Two further arguments bear on the second part.

- Hyperlink–citation semantics: In the original PageRank algorithm (Page et al., 1999), an inbound hyperlink counts as a citation. H.-D. Do and Do (2023) apply this reasoning to Ethereum transfers, and AWP (T. Do et al., 2023) weights each transfer edge by the age and the value of its transfers, through a logistic decay in age and a bounded transform of value (Section 3.6.1). EndorseRank extends citation-based propagation to ERC-20 approve and permit edges. An allowance is an explicit, on-chain delegation of spending authority that remains in force until it is spent, revoked or replaced (Vogelsteller & Buterin, 2015), whereas a transfer may reflect trading or routing without any intent to trust. Graph-based risk models propagate risk through the transaction network to rate accounts that carry no label (D. Lin et al., 2025); whether a directed graph of authorizations adds to such propagation over transfers is the question taken up here.
- Domain alignment framework: Measurement theory uses construct alignment to ask how closely an indicator follows the construct it is meant to track (Cronbach & Meehl, 1955). Messick (1995) widens the idea into one unified judgment about what a score means and what using it implies. Graph coherence is checked with rank correlations (Section 2.9) against contemporaneous transfer and allowance statistics. The out-of-window checks are temporal holdouts: new approvals and new transfer senders for spenders, and, in the registered July–September window W1, the liquidation-free close rate for traders.
- Computational complexity: Each PageRank iteration costs $O(|E|)$ (Section 2.5.2), so on the sparser allowance graph an iteration needs less time and memory; this argument underlies H3.

The number of iterations is a separate factor: the damping factor bounds its rate of convergence (Langville & Meyer, 2006), but the count also depends on the graph and the restart vector. Both methods use the same solver, so a difference in runtime comes from the graphs and the restart vectors and does not count as an algorithmic contribution (Appendix 8.3.3).

- Local counts as a baseline: Where contracts that already have many counterparties gain the most new ones (Section 2.4), the raw degree is a strong predictor of future arrivals. A PageRank differs from it in three ways: the endorsers' own scores propagate along the edges, the edges are weighted by age and amount, and the walk restarts by a rule of its own; a contrast with the degree tests the three together. RQ4 therefore sets each PageRank against its own raw degree at the freeze date and does not credit a difference to propagation alone.
- Layer coupling: Multiplex PageRank (Halu et al., 2013) and multilayer ranking (De Domenico et al., 2015) treat several relations as layers over one node set. Mixing the layers node by node, as C-PR does, leaves a question open for layers of very different density: a node with no edge in the sparse layer follows the dense one, so the signal of the sparse layer may or may not survive. RQ5 therefore asks two things: whether the mixture improves on its transfer layer walked alone, and whether it beats both layers. Both layers weight edges by US-dollar amounts without AWP's bounded value weight and are walked with uniform restarts, so neither walk is AWP or EndorseRank, and C-PR is compared with AWP separately (Section 4.6.4). Coverage matters there: a transfer walk leaves all nodes without transfer edges in one tie, and any walk over the allowance edges, EndorseRank included, orders them, so coupling can add coverage over a transfer walk even if it adds nothing where both layers reach.

The conceptual model in Figure 2.3 links the scores of the thesis to H1–H3 and RQ1–RQ4. RQ5, the dashed box on the right, takes input from H2/RQ2 and from RQ4 and asks whether the two constructs combine.

Only EndorseRank and AWP enter H1. H2/RQ2 and H3/RQ3 concern every score, and RQ4 is a time-split test on the spender cohort; it does not rank trading outcomes.

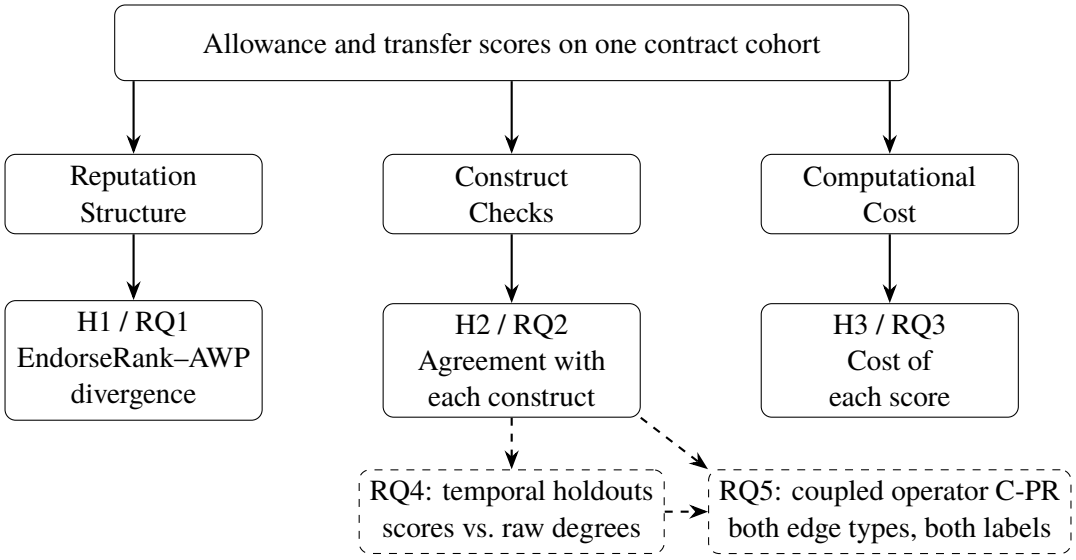


Figure 2.3: Conceptual model linking the allowance and transfer scores of one contract cohort to reputation structure (H1/RQ1), construct checks (H2/RQ2), computational cost (H3/RQ3) and the temporal holdout, in which each PageRank is set against the raw degree it smooths (RQ4). The dashed box on the right is RQ5, the coupled operator C-PR.

2.11 Terminology

The formal constructs are defined in Chapter 1. The terms below are used when results are interpreted:

- Single-layer methods: EndorseRank (allowance graph) and AWP (transfer graph), the two PageRank scores that walk on one edge type. The coupled walks of Section 3.6.3 and the raw degrees are evaluated beside them.
- Primary graph construct: the graph that a method is *designed* to rank, for example G_{approve} for EndorseRank, as distinct from the checks applied after scoring.
- Contemporaneous check: a local transfer or allowance statistic taken from the same window as the score (Cronbach & Meehl, 1955).
- Temporal holdout: in the first window, W0, scores are frozen at t_1 (31 March 2026), and new owner–spender approval pairs and new transfer senders are counted up to t_2 (30 June 2026); a registered second window, W1, freezes scores on 30 June 2026 and counts labels from July

to September 2026 (Section 3.7.2).

- Matched contract cohort, spender holdout cohorts of W0 and W1, trader cohorts and subsamples of the matched cohort: the address sets of Section 4.1.1.

2.12 Domain alignment metrics

Construct alignment is measured by Spearman's ρ and Kendall's τ between scores and check rankings (Section 2.9), on contemporaneous transfer and allowance checks and on the holdout labels; Chapter 3 gives the operational definitions.

2.13 Worked example: PageRank on a four-node endorsement graph

Take four addresses $\{A, B, C, D\}$ joined by the latest-allowance edges

$$A \to B\ (2), \quad A \to C\ (1), \quad B \to D\ (3), \quad C \to D\ (1),$$

with damping $d = 0.85$. The out-strengths are $o_A = 3$, $o_B = 3$, $o_C = 1$ and $o_D = 0$ (the last is dangling). The edges $B \to D$ and $C \to D$ are the only out-edges of their nodes, so each is followed with probability one and its weight has no effect on the scores. Teleportation adds $(1 - d)/4 = 0.0375$ to every node at every step, the first one included. One power-iteration step from the uniform vector $\mathbf{r}^{(0)} = \mathbf{1}/4$ gives $\mathbf{r}^{(1)} = (0.0906,\ 0.2323,\ 0.1615,\ 0.5156)$ for (A, B, C, D). The mass already tilts toward D, since both B and C endorse D, while A splits its mass between B and C. The iteration converges to the stationary vector $\mathbf{r} = (0.1375,\ 0.2154,\ 0.1765,\ 0.4706)$, so D comes first, B and C follow, and A is last. No address endorses A. Apart from teleportation, its whole score comes from the dangling rule, which spreads D's mass evenly over the four nodes ($d\, r_D/4 = 0.1000$). The example shows that a node gains score by *receiving* endorsements from nodes that are endorsed in turn, which is precisely the citation logic of PageRank (Langville & Meyer, 2006; Page et al., 1999).

Suppose the same addresses were linked by transfers instead of allowances. The transfers would in general join other pairs, or the same pairs with other per-pair totals. That gives another transition matrix P, and possibly a different ranking of the same nodes. Hypothesis H1 is this situation in miniature: EndorseRank and AWP need not agree. They also restart differently, so H1 compares edges and restarts together; Section 4.5.4 walks the allowance graph with AWP's restarts.

2.14 DeFi risk taxonomy and where reputation fits

Werner et al. (2022) separate the technical security of DeFi protocols from their economic security, and Zhou et al. (2023) place the attacks and accidents seen in practice in one reference frame. Bugs in contract code are smart-contract risks, one of the several levels at which DeFi vulnerabilities arise (Li et al., 2022). A stale price, or a feed under someone's control, is an oracle risk (Angeris & Chitra, 2020). Market risks (Perez et al., 2021) cover volatility and liquidation cascades, including stress in stablecoin and funding markets (Klages-Mundt et al., 2020), and the market crash of March 2020 showed how these two layers can interact (Gudgeon, Perez, et al., 2020). Token-weighted voting belongs to governance risk: in the on-chain governance of Compound, Uniswap and ENS, a small number of addresses hold most of the voting power (Fritsch et al., 2024). We study a narrower layer, *counterparty* or *behavioral* risk: whether an address's past behavior makes it a safe counterparty, as a trader that handles leveraged exposure responsibly or as a contract that many owners trust to spend their tokens.

On-chain reputation scores cover only part of the behavioral risk layer and are no substitute for collateral, auditing, or oracle design. Their purpose is to compress a system's relational structure (who authorizes whom, who pays whom) into an ordinal measure that others can audit and reproduce from public data. Within behavioral risk, EndorseRank looks at authorization and AWP at payment flows. Neither tries to predict exploits or governance attacks.

Finally, fake activity makes behavioral risk harder to measure. Wash trading inflates volume on decentralized exchanges (Victor & Weintraud, 2021) and, on a larger scale, on unregulated centralized

exchanges (Cong et al., 2023). Front-running and related forms of extractable value create transfers that follow ordering strategies instead of relationships (Daian et al., 2020); Qin et al. (2022) estimate how much of that value is captured on Ethereum. Scam tokens on Uniswap (Xia et al., 2021) likewise add transactions without forming sustained relationships. The transfer graph is particularly sensitive to fake volume, while the allowance graph is vulnerable to approval farming. Time-based splits and Sybil detection solve the problem only in part.

2.15 Related measurement traditions

Outside blockchains, reputation systems have been studied in e-commerce (Resnick & Zeckhauser, 2002), in peer-to-peer networks (Kamvar et al., 2003), and in work on web spam (Gyöngyi et al., 2004). Four insights from this work apply here: (i) reputation is a property of relations; (ii) when identities are easy to create, Sybil attacks become possible (Douceur, 2002); (iii) seeding and normalization limit the effect of inflation; and (iv) evaluation needs external benchmarks, because internal coherence alone is not enough (Cronbach & Meehl, 1955). EndorseRank takes up (i) directly and treats (ii) as a threat. Of (iii) it takes up normalization, since each owner's endorsement is divided over the spenders it approves, but not seeding, since it restarts uniformly (Section 2.6). On (iv) it goes part of the way: the contemporaneous checks, measured by Spearman's ρ (Spearman, 1904) and Kendall's τ (Kendall, 1938), test coherence, while the temporal holdouts test prediction out of window, on the same constructs for spenders and on liquidation for traders.

Personalized and topic-sensitive PageRank (Haveliwala, 2003; Jeh & Widom, 2003) bias the teleportation vector toward a chosen group of nodes, the mechanism that TrustRank uses to turn a small seed set into a global ranking (Gyöngyi et al., 2004); AWP's restarts in proportion to sending activity are another instance (T. Do et al., 2023). Multiplex PageRank lets a node's centrality on one layer shape the walk on another (Halu et al., 2013), and the multilayer framework of De Domenico et al. (2015) lets one walker move within and between the layers of an interconnected network. Both treat layers as different relations over one node set, which matches allowances and transfers

among the same addresses. C-PR takes the simplest form of this idea, a per-node weighted sum of two transition matrices with one weight, and S-PR borrows the personalized-teleportation idea, with the allowance score as the teleportation vector (Section 3.6.3).

2.16 Accounts, gas, and why logs are the measurement surface

Ethereum has two kinds of account: externally owned accounts (EOAs) and contract accounts. A private key controls an EOA, whereas a contract account is governed by the code it stores (Antonopoulos & Wood, 2018). This thesis scores contract accounts only (Section 3.2). Every state-changing transaction consumes gas. Logs come from a contract's `LOG` instructions, which run when its code emits an event (the Solidity `emit` statement); writing to storage produces no log. A reputation system that means to be auditable must prefer logs to off-chain databases, because logs are a permanent part of the blockchain history that any full node or index provider can recompute independently (Wood, 2025).

Transfers recur routinely, while an approval is granted once so that later transfers can go through, so G_{transfer} is expected to be the denser graph, and low L2 fees may widen the gap; Chapter 4 reports the resulting graph sizes.

2.17 Approvals as security-relevant events

Unlimited approvals (`type(uint256).max`) save the owner from approving again, at a cost in safety. If the approved account is compromised, every one of the owner's tokens can be stolen until the owner resets the approval. An allowance is therefore a capability and not a record: whoever holds it can move the owner's tokens until it is spent or revoked (Vogelsteller & Buterin, 2015). Transaction-based phishing exploits this: fake websites lead users to sign transactions that let scammers take their assets (He et al., 2023). EndorseRank interprets a token approval as endorsing one spender's capability to move those tokens. An account could receive many transfers and never be granted permission to move funds on anyone else's behalf.

Wang et al. (2022) find that unlimited approvals make up the majority of ERC-20 approvals on Ethereum and quantify the risk they carry. Here they are close to half of the positive latest allowances (Appendix 8.4.2 gives the counts) and appear as amounts of $2^{256} - 1 \approx 1.16 \times 10^{77}$ base units. EndorseRank's value weight gives them the full weight of one, which a finite allowance approaches only at a few times the median amount, while in the allowance layer of C-PR, which counts every allowance up to a cap in US dollars (Section 3.6.3), an unlimited approval weighs no more than a large finite one. Just over half of the owners hold at least one.

2.18 AWP in depth: what the transfer-graph baseline measures

Adaptive Weighted PageRank (AWP), proposed by T. Do et al. (2023), ranks the addresses of a transaction graph. Each transaction adds to its edge a logistic function of its time multiplied by a bounded function of its value, so that no transaction adds more than one to the edge weight. The walk restarts at each address in proportion to its activeness, the sum over the addresses it has sent to of the time weight of its latest transaction to each. Ranking accuracy is then validated against inbound degree and inbound value, taken as proxies for the economic attention a wallet attracts. The form of the decay (Equation (3.3)) and the in-degree/in-value proxies in Chapters 3 and 4 are taken from the AWP paper. The paper gives no parameter values; Section 3.6.1 gives the values used here. The approach is entirely on-chain and rank-based and needs no off-chain identity.

EndorseRank keeps AWP's weight functions under PageRank and its rank-correlation check, and changes what the edges mean and where the walk restarts: EndorseRank restarts uniformly, AWP at active senders (Section 3.6.2). In Chapter 4 we test whether allowance edges yield a score that is construct-distinct, allowance-aligned, and cheaper, and whether scores taken at t_1 can predict new approvals by t_2.

2.19 Comparison of graph reputation methods

Table 2.1 compares EndorseRank with the other approaches covered in this chapter by edge semantics, temporal model and validation design.

Table 2.1: Comparison of selected graph reputation methods (conceptual).

Method	Edge semantics	Temporal model	Typical validation
PageRank (web) (Page et al., 1999)	Hyperlinks	None / static snapshot	Retrieval quality
EigenTrust (Kamvar et al., 2003)	Local trust ratings	Iterative aggregation	P2P file authenticity
TrustRank (Gyöngyi et al., 2004)	Links from seed goods	Static / seeded	Spam demotion
PageRank/ HodgeRank on Ethereum (H.-D. Do & Do, 2023)	Token transfers	Static window	Not reproduced in this thesis
AWP (T. Do et al., 2023)	Transactions, bounded value weights	Logistic decay of edges and restarts	In-degree / in-value
RiskProp (D. Lin et al., 2025)	Transactions, with a de-anonymization score	Propagation of risk	Labeled illicit accounts
Multiplex / multilayer PageRank (De Domenico et al., 2015; Halu et al., 2013)	Several relations over one node set	Static layers	Versatility of top-ranked nodes
Human Passport (n.d.)	No edges; weighted credentials about the address holder, on-chain and off-chain	Current credential set	Admission threshold; known human and Sybil addresses
EndorseRank (this study)	Latest allowances of five tokens in US dollars, AWP's weight functions	Logistic decay of each latest approval; uniform restarts	Allowance checks; t_2 new approvals
C-PR / S-PR (this study)	Allowances mixed with, or seeding, transfers; US-dollar amounts, allowances capped	Allowance snapshot and logistic decay of transfers	Holdout labels; rules fixed in advance, re-applied to the revised scores

As Table 2.1 indicates, EndorseRank's contribution is the use of the latest allowance as the edge of

a reputation ranking, evaluated out of window; it is neither the ranking algorithm nor the reading of approvals as a link between addresses (Section 2.7). On the operator side, our contribution is to couple the two relations into one walk, with a seeded ablation to measure how much this coupling contributes. The coupling is tested out of window and against both constructs, under rules fixed before this study extracted the data; on the revised scores of the main analysis they are applied as a re-analysis (Section 3.5.1 and Appendix 8.6).

2.20 Summary of the reviewed literature

Table 2.2 collects the sources underlying this study, with the strengths this thesis draws on and the limitations it tries to address.

Table 2.2: Summary of the reviewed literature: methodology, strengths, limitations, and relation to this study.

Source (year)	Methodology	Strengths	Limitations	Relation to this study
Page et al. (1999); Langville and Meyer (2006)	Eigenvector ranking of a hyperlink graph, solved by power iteration	Interpretable propagation; $O(\|E\|)$ per iteration; damping yields a unique Perron vector	Edges can only mean hyperlinks; no notion of time	Same solver and complexity argument, used unchanged for both methods compared
Kamvar et al. (2003); Gyöngyi et al. (2004)	Aggregation of trust in P2P networks; propagation from seeds to demote spam	Normalization and seed sets curb inflation; the adversary is modeled explicitly	Depends on explicit ratings or curated seeds; not applied on-chain	Informs the Sybil-cost discussion and the uniform-teleportation baseline
H.-D. Do and Do (2023)	PageRank and HodgeRank applied to Ethereum transaction graphs as a measure of social credit	Treats transfer connectivity as relevant to credit; data are fully public	Transfers conflate payment with trust; allowance semantics absent	Earlier transfer-graph work, kept separate from AWP in this thesis

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Table 2.2 (continued)

Source (year)	Methodology	Strengths	Limitations	Relation to this study
T. Do et al. (2023), AWP	Transaction PageRank with bounded value weights, logistic recency and restarts weighted by sending activity; rank correlation with in-degree/in-value as validation	Validates on ranks, so no default labels are needed; recency modeled explicitly	Proxies are computed from the same transfer data that forms the graph (mechanical alignment); dense graph; no parameter values reported; contracts without transfer edges all tie at zero	Baseline, reimplemented from the published formulas with k, t_0 and b set here (Section 3.6.1); EndorseRank takes the same weight functions with its own slope
D. Lin et al. (2025), RiskProp	Network propagation of account risk, paired with a deanonymization score	Rates every account, including those that clustering and classification leave unassessed; checked against labeled accounts	Checked against illicit-account labels; aimed at fraud, not trust or authorization	Shows that propagation over transactions adds to per-account methods; not reproduced here, since this study uses no fraud labels
Wu et al. (2022)	Phishing detection through network embedding of Ethereum transaction graphs	Embedding that uses the amounts and times of transactions; more effective than earlier embedding methods	Needs labeled phishing addresses; its labels mark phishing, not creditworthiness	Backs the premise that graph position is informative; not a competing method
Hassija et al. (2020); Kotb (2023)	Blockchain-data credit scoring using prospect theory and machine learning	Aimed directly at lending decisions	Need labels or off-chain features; only partly interpretable	A contrasting class; EndorseRank remains within auditable graph propagation

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Table 2.2 (continued)

Source (year)	Methodology	Strengths	Limitations	Relation to this study
Bellini et al. (2020); Almasoud et al. (2020)	Surveys of blockchain-based trust and reputation systems	Taxonomies of the design space; a review of thirty studies of reputation systems built on smart contracts	Reputation taken from users' ratings of services; authorization edges not considered; nothing measured on L2	None of the surveyed designs uses latest-allowance edges
Victor (2020); Victor and Weintraud (2021); Cong et al. (2023)	Heuristics for clustering addresses, one of which links addresses through ERC-20 approvals; detection of wash trading on DEXs and exchanges	Measure the share of on-chain activity that is artificial or multi-address	No reputation score proposed	Closest prior use of approvals, as a clustering link and not a ranking edge; show why transfer-based signals are open to inflation
Cronbach and Meehl (1955); Messick (1995)	Psychometric construct validity	Distinguish what a score is designed to measure from external criteria	Not specific to blockchains	Basis for treating the contemporaneous checks as construct checks and the temporal holdouts as out-of-window evidence
Human Passport (n.d.); Siddarth et al. (2020); Bordeianu and Popescu (2026)	Weighted credentials about an address holder, on-chain and off-chain, against a passing threshold; reviews of proof-of-personhood defenses	In deployed use; adds off-chain evidence against Sybil attacks	A gate and not a graded ordering; relations between addresses play no part; Sybil strategies keep changing	Shows how trust is established in practice, and what EndorseRank does not offer: no personhood guarantee

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Table 2.2 (continued)

Source (year)	Methodology	Strengths	Limitations	Relation to this study
Efron (1979); DiCiccio and Efron (1996)	Bootstrap resampling and bootstrap-based confidence intervals	Uncertainty for rank statistics with no distributional assumption; paired contrasts	Units must be exchangeable	Source of the 95% intervals and $\Delta\tau$ contrasts reported in Chapter 4

2.21 Research gap

Five gaps emerge from Table 2.2 and Section 2.15. First, every on-chain reputation method discussed here (PageRank/HodgeRank (H.-D. Do & Do, 2023), AWP (T. Do et al., 2023), RiskProp (D. Lin et al., 2025), and the designs in the surveys (Bellini et al., 2020)) builds its graph from transfers or explicit ratings, and the attestation tooling deployed in DeFi scores the holder of an address without any graph of relations between addresses (Human Passport, n.d.). None uses the ERC-20 allowance, the on-chain record of a delegation of spending power that every ERC-20 token carries. Approvals have served address clustering and security analysis (Sections 2.7 and 2.17), but in the work reviewed here they have not served as the edges of a reputation ranking. It remains unknown whether a latest-allowance endorsement graph yields a distinct and meaningful ordering (O1 and RQ1). A second gap lies in validation. The transfer-graph baseline checks its scores against in-degree and in-value computed from the same transfer data used to build the graph, so some agreement is unavoidable; no report applies this procedure to an allowance construct and its transfer counterpart with confidence intervals and paired difference tests on the same set of addresses (O2 and RQ2). Third, the transfer-graph studies reviewed here give no runtimes with edge counts, memory use and iteration counts under a fixed protocol, which makes it hard to attribute any efficiency claim to sparsity as opposed to implementation (O3 and RQ3). The fourth gap is methodological and has to do with time. Graph statistics from the same window are often presented as validation, although such scores cannot be shown to predict future behavior; the out-of-window

check used here is instead a temporal holdout with a future-tense label for each construct and raw degree at the time of the freeze as the baseline (O4 and RQ4). Finally, multiplex PageRank (Halu et al., 2013) and multilayer ranking (De Domenico et al., 2015) have been applied to social networks and to other networks with several kinds of relation, but, in the work reviewed here, not to two on-chain relations over one set of addresses. No study reports whether combining an authorization layer with a payment layer improves on the payment layer alone, or beats both layers, under rules fixed before the outcome is seen (O5 and RQ5).

2.22 Open problems motivating this research

The literature leaves several problems in on-chain reputation unresolved, and these motivate the present study:

1. Edge semantics: When identities are pseudonymous (D. Lin et al., 2024), which on-chain relations capture trust or authorization best (Packin & Lev-Aretz, 2024)?
2. Validation without defaults: On what basis can methods be compared when loan-default labels are scarce (H.-D. Do & Do, 2023)?
3. Computational cost: As graphs grow, can reputation scores still be refreshed at interactive latencies (Langville & Meyer, 2006)?
4. Time-split checks: What keeps an evaluation from mistaking same-window graph statistics for out-of-window evidence (Cronbach & Meehl, 1955)?
5. Adversarial robustness: How much does it cost to forge reputation when identities are cheap (Douceur, 2002)?

EndorseRank takes on (1)–(4) through allowance edges, contemporaneous checks, runtime benchmarks, and a temporal holdout. Problem (5) is analyzed at the model level in Chapter 5; a full empirical adversarial evaluation is left for future work.

Chapter 3 turns this gap statement into an operational design.